

Администрирование локальных сетей

Статическая маршрутизация VLAN в Cisco Packet Tracer

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Цель работы

Настроить статическую маршрутизацию VLAN в сети и изучить процесс передачи пакетов между виртуальными локальными сетями в Cisco Packet Tracer.

Построение сетевой топологии

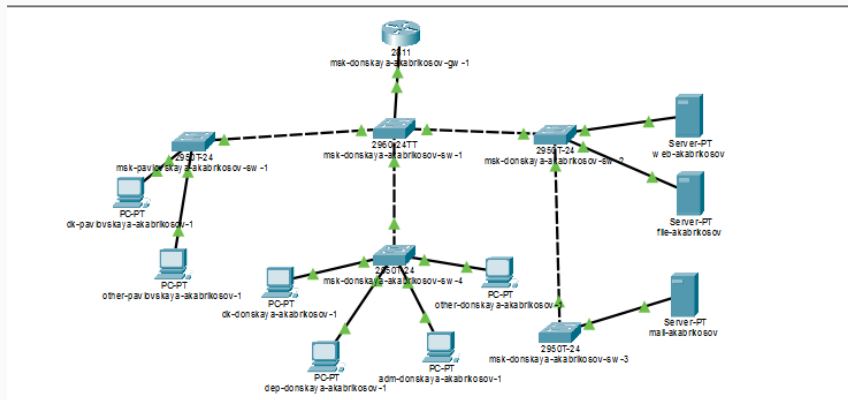


Рис. 1: Топология сети с маршрутизатором и VLAN

Первоначальная конфигурация

```
Router>en
Router#conf term
      ^
% Invalid input detected at '^' marker.

Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-akabrikov-gw-1
msk-donskaya-akabrikov-gw-1(config)#line vty 0 4
msk-donskaya-akabrikov-gw-1(config-line)#pass cisco
msk-donskaya-akabrikov-gw-1(config-line)#login
msk-donskaya-akabrikov-gw-1(config-line)#
msk-donskaya-akabrikov-gw-1(config-line)#line console 0
msk-donskaya-akabrikov-gw-1(config-line)#pass cisco
msk-donskaya-akabrikov-gw-1(config-line)#login
msk-donskaya-akabrikov-gw-1(config-line)#
msk-donskaya-akabrikov-gw-1(config-line)#enable secret cisco
msk-donskaya-akabrikov-gw-1(config)#
msk-donskaya-akabrikov-gw-1(config)#serv pass
msk-donskaya-akabrikov-gw-1(config)#
msk-donskaya-akabrikov-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-akabrikov-gw-1(config)#ip domain-name donskey.rudn.edu
msk-donskaya-akabrikov-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-akabrikov-gw-1.donskey.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-akabrikov-gw-1(config)#line vty 0 4
*Mar 1 0:4:9.6: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:4:9.6: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-akabrikov-gw-1(config-line)#trans in ssh
msk-donskaya-akabrikov-gw-1(config-line)#exit
msk-donskaya-akabrikov-gw-1(config)#
```

Конфигурация подинтерфейсов

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
msk-donskaya-akabrikosov-gw-1(config-subif)#encapsulation dot1q 101
msk-donskaya-akabrikosov-gw-1(config-subif)#ip addr 10.128.3.1 255.255.255.0
msk-donskaya-akabrikosov-gw-1(config-subif)#desc dk
msk-donskaya-akabrikosov-gw-1(config-subif)#
msk-donskaya-akabrikosov-gw-1(config-subif)#int f0/0.102
msk-donskaya-akabrikosov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up
msk-donskaya-akabrikosov-gw-1(config-subif)#encapsulation dot1q 102
msk-donskaya-akabrikosov-gw-1(config-subif)#ip addr 10.128.4.1 255.255.255.0
msk-donskaya-akabrikosov-gw-1(config-subif)#desc departaments
msk-donskaya-akabrikosov-gw-1(config-subif)#
msk-donskaya-akabrikosov-gw-1(config-subif)#int f0/0.103
msk-donskaya-akabrikosov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.103, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
msk-donskaya-akabrikosov-gw-1(config-subif)#encapsulation dot1q 103
msk-donskaya-akabrikosov-gw-1(config-subif)#ip addr 10.128.5.1 255.255.255.0
msk-donskaya-akabrikosov-gw-1(config-subif)#desc adm
msk-donskaya-akabrikosov-gw-1(config-subif)#
msk-donskaya-akabrikosov-gw-1(config-subif)#encapsulation dot1q 104
msk-donskaya-akabrikosov-gw-1(config-subif)#encapsulation dot1q 104
msk-donskaya-akabrikosov-gw-1(config-subif)#ip addr 10.128.6.1 255.255.255.0
msk-donskaya-akabrikosov-gw-1(config-subif)#desc other
msk-donskaya-akabrikosov-gw-1(config-subif)#exit
msk-donskaya-akabrikosov-gw-1(config)#exit
msk-donskaya-akabrikosov-gw-1#
%SYS-5-CONFIG I: Configured from console by console
```

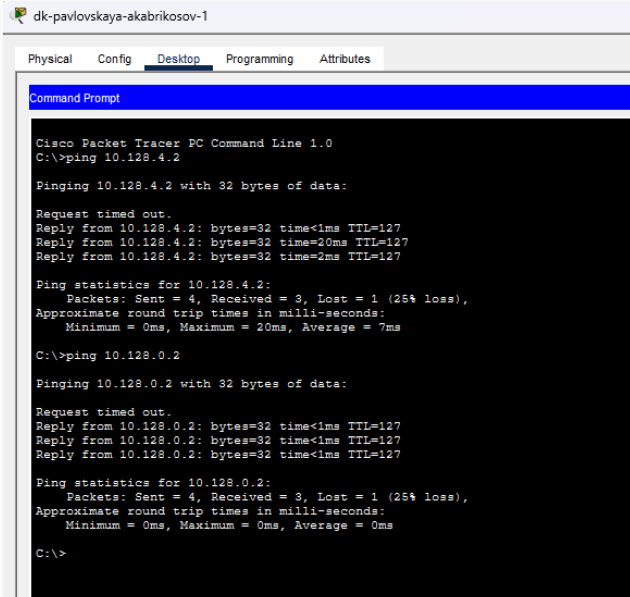
msk-donskaya-akabrikosov-gw-1

Physical Config CLI Attributes

IOS Command Line Interface

```
!
!
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.2
description management
encapsulation dot1Q 2
ip address 10.128.1.1 255.255.255.0
!
interface FastEthernet0/0.3
description servers
encapsulation dot1Q 3
ip address 10.128.0.1 255.255.255.0
!
interface FastEthernet0/0.101
description dk
encapsulation dot1Q 101
ip address 10.128.3.1 255.255.255.0
!
interface FastEthernet0/0.102
description departaments
encapsulation dot1Q 102
ip address 10.128.4.1 255.255.255.0
!
interface FastEthernet0/0.103
description adm
encapsulation dot1Q 103
ip address 10.128.5.1 255.255.255.0
!
interface FastEthernet0/0.104
description other
encapsulation dot1Q 104
ip address 10.128.6.1 255.255.255.0
!
interface FastEthernet0/1
--More--
```


Проверка соединения



The screenshot shows a Cisco Packet Tracer PC Command Line interface. The window title is 'dk-pavlovskaya-akabrikosov-1'. The 'Desktop' tab is selected. The Command Prompt displays the output of two ping commands. The first command is 'ping 10.128.4.2', which shows a 'Request timed out' followed by three successful replies with times of <1ms, 20ms, and 2ms. The second command is 'ping 10.128.0.2', which also shows a 'Request timed out' followed by three successful replies with times of <1ms. Both commands include statistics showing 4 packets sent, 3 received, and 1 lost (25% loss).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.4.2

Pinging 10.128.4.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.2: bytes=32 time<1ms TTL=127
Reply from 10.128.4.2: bytes=32 time=20ms TTL=127
Reply from 10.128.4.2: bytes=32 time=2ms TTL=127

Ping statistics for 10.128.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 7ms

C:\>ping 10.128.0.2

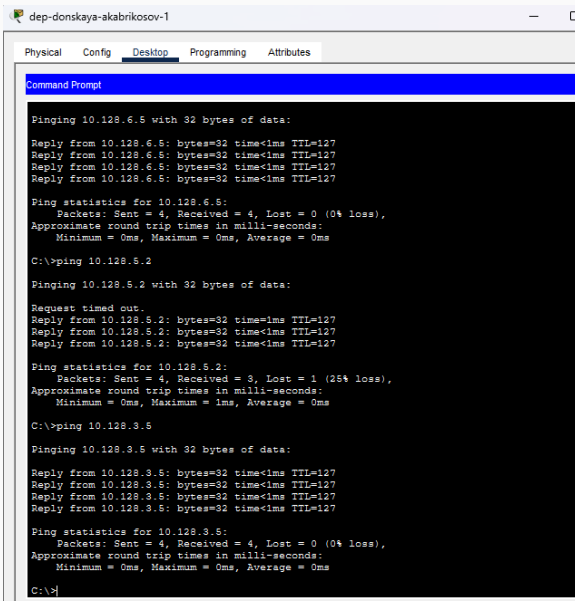
Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Тестирование соединения



```
dep-donskaya-akabrikosov-1
Physical Config Desktop Programming Attributes
Command Prompt

Pinging 10.128.6.5 with 32 bytes of data:

Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.2

Pinging 10.128.5.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.2: bytes=32 time=1ms TTL=127
Reply from 10.128.5.2: bytes=32 time<1ms TTL=127
Reply from 10.128.5.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.3.5

Pinging 10.128.3.5 with 32 bytes of data:

Reply from 10.128.3.5: bytes=32 time<1ms TTL=127
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Анализ передачи пакетов

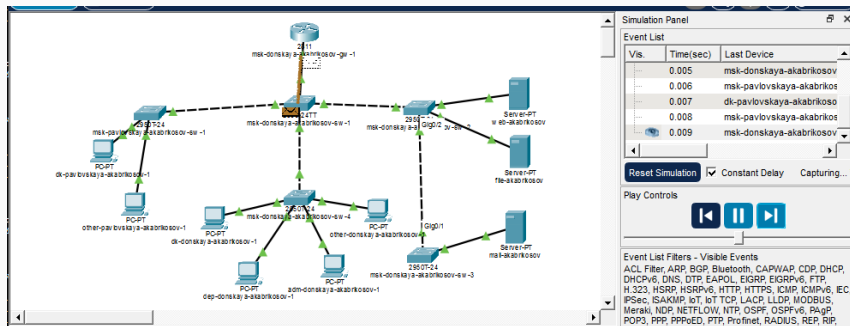


Рис. 7: Передвижение пакета в режиме Simulation

PDU Information at Device: msk-donskaya-akabrikov-sw-1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

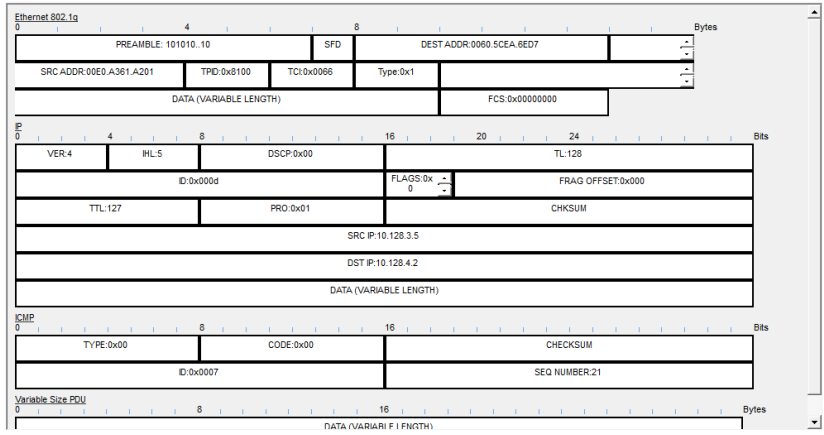


Рис. 8: Структура передаваемого пакета

Вывод

В ходе лабораторной работы:

- настроен маршрутизатор Cisco 2811
- реализована маршрутизация между VLAN
- выполнена настройка trunk-соединения
- исследована передача пакетов ICMP